UNIVERSITY OF LEICESTER

A candidate-variant association study of self-reported asthma in mothers of the Cebu Longitudinal Health and Nutrition Survey (CLHNS) cohort

Alvin E. Lirio[1], K Alaine Broadaway[2], Jing Chen[1], Paulita Duazo[3], Karen Mohlke[2], Nanette Lee[3], Catherine John[1], Linda Adair[2], Martin D. Tobin[1,4], Anna L. Guyatt[1]

[1]Department of Population Health Sciences, University of Leicester, Leicester, United Kingdom; [2]UNC Chapel Hill Genetics Department, Chapel Hill, USA; [3]USC Office of Population Studies, University of San Carlos, Cebu City, Philippines; [4]Leicester NIHR Biomedical Research Centre, Leicester, United Kingdom

Background

Asthma is an obstructive respiratory disease for which morbidity is greatest in low- and middle-income countries: 15% of the Filipino population are estimated to have asthma, half of whom have inadequately controlled disease.[1]

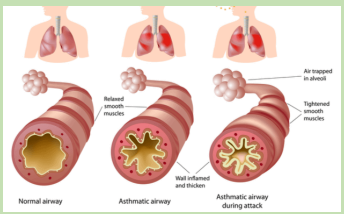


Figure 1. Pathophysiology of asthma

Asthma is characterised by reversible airflow obstruction due to airway narrowing (bronchoconstriction), airway wall thickening, and increased mucous secretion.

Factors such as exposures to polluted air, respiratory infections, tobacco smoking, and a family history of asthma or allergies, have been proposed as potential causes for asthma development.



Figure 2. Risk factors for asthma

A substantial number of genetic variants for asthma have been identified,[2] but a lack of ancestry representation means that understudied populations do not yet stand to benefit equally from genomic research.[3] This study aims to perform the first genetic association study of asthma in a Filipino population.

Methods

Study population

1,648 unrelated mothers (129 with self-reported asthma) of the Cebu Longitudinal Health and Nutrition Survey (CLHNS) were included in the analysis. CLHNS is an interdisciplinary longitudinal population birth cohort study based in Metropolitan Cebu in the Philippines, comprising of broad collection of information including demographics, socio-economics, and health conditions of mothers, offspring, and household.

Candidate-variant association study

Using the largest published multi-ancestry asthma GWAS (Global Biobank Meta-analysis Initiative, GBMI (153,763 cases and 1,647,022 controls)),[2] we selected sentinel single-nucleotide polymorphisms (SNPs) associated at $P<5x10^{-8}$ in the multi-ancestry analysis, and $P<5x10^{-6}$ in GBMI East Asian (EAS) participants (18,549 cases and 322,655 controls).

Genetic Risk Scores (GRS) analysis

Genetic variants were aggregated into a genetic risk score (GRS), weighted by the EAS GBMI asthma GWAS.[2]

Details of genetic association testing

- Associations with asthma were tested using logistic regression, with SNPs (or proxies r2>0.8) coded additively, adjusting for age and 15 principal components
- Principal components were calculated using EIGENSOFT
- Power calculations done using QUANTO
- genome-wide significance set at $P<5x10^{-8}$

Results

Candidate-variant association study

Twenty-seven candidate SNPs were analysed. An intronic variant in SMAD3 (rs17293632, OR 1.80, 95%CI:1.28-2.54, P=0.0008) showed evidence of association with asthma in CLHNS mothers. SMAD3 is involved in the TGF-β signalling pathway, which is related to airway remodelling in asthma.[5] No other variant passed correction for 27 tests.

Table 1. Candidate variant association analysis in self-reported asthma.

CHR	POS	RSID	Effect Allele	EAF	Beta	SE	*p*-value
1	226909233	1:226909233:A:G	A	0.1618	-0.1591	0.1841	0.3876
2	8441735	2:8441735:C:G	G	0.2291	-0.3126	0.1687	0.0639
2	24945954	2:24945954:A:G	G	0.0989	0.4310	0.1893	0.0228
2	102931534	2:102931534:C:T	T	0.0140	-1.2742	1.0155	0.2096
2	204584456	2:204584456:A:T	T	0.2768	0.1020	0.1449	0.4816
2	242700770	2:242700770:A:G	G	0.0074	0.5119	0.6297	0.4162
3	33047662	3:33047662:G:T	T	0.3310	-0.0999	0.1416	0.4805
4	123132492	4:123132492:A:G	G	0.2193	0.1237	0.1475	0.4017
5	110407507	5:110407507:C:T	T	0.2434	0.0719	0.1495	0.6304
5	131995964	5:131995964:A:G	A	0.3942	-0.0853	0.1357	0.5296
5	141494934	5:141494934:C:T	C	0.4920	-0.0747	0.1327	0.5737
6	32572202	6:32572202:C:T	T	0.2017	0.1313	0.1562	0.4005
7	3146081	7:3146081:C:T	T	0.3781	-0.0109	0.1335	0.9352
7	20557179	7:20557179:A:G	G	0.3912	0.1016	0.1331	0.4453
8	129399732	8:129399732:C:T	T	0.1648	-0.1339	0.1847	0.4684
9	92226172	9:92226172:C:G	G	0.3561	-0.1664	0.1404	0.2358
10	9061370	10:9061370:C:T	C	0.1085	0.2126	0.1945	0.2743
10	64396916	10:64396916:C:T	T	0.4339	-0.1589	0.1311	0.2255
11	76292573	11:76292573:A:T	T	0.3114	0.0275	0.1425	0.8472
15	61066516	15:61066516:C:T	T	0.2354	-0.2850	0.1644	0.0829
15	***67442596***	***15:67442596:C:T***	***T***	***0.1156***	***0.5889***	***0.1754***	***0.0008***
16	27354531	16:27354531:C:G	C	0.4213	-0.0388	0.1358	0.7754
17	7786699	17:7786699:G:T	T	0.1097	-0.2967	0.2350	0.2067
17	38063381	17:38063381:C:T	T	0.2443	-0.1302	0.1549	0.4008
17	47468011	17:47468011:A:G	G	0.3421	-0.1360	0.1424	0.3395
18	45473449	18:45473449:C:T	T	0.2217	0.2185	0.1508	0.1474
19	9129660	19:9129660:C:T	C	0.4717	-0.0953	0.1325	0.4723

CHR – chromosome, POS – base-pair position, RSID – chromosome number:position_effect allele_alternate allele; EAF – effect allele frequency; Beta – log odds ratio; SE – standard error of the log odds ratio

Genetic Risk Score analysis

The GRS was associated with increased odds of asthma (OR per weighted allele increase 1.07 [95%CI:1.01-1.14], P=0.022).

Conclusion

Despite the small sample size and number of asthma cases in CLHNS, we detected a known asthma genetic variant using a candidate-variant approach, and demonstrated the value of aggregating variants into risk scores (with confidently ascribed weights from a large published GWAS) when power is limited. However, we were underpowered to undertake a GWAS, and thus to discover novel asthma variants in a Filipino population. Commitment to cohort development and investment in understudied populations by the genomics field as a whole will be key to addressing inequity in asthma genetic research.

References

1. Ho, F. D. V. et al. Lancet Respir Med 11, 513-514, doi:10.1016/s2213-2600(23)00180-7 (2023).
2. Tsuo, K. et al. Cell Genom 2, 100212, doi:10.1016/j.xgen.2022.100212 (2022).
3. Bien, S. A. et al. Annu Rev Genomics Hum Genet 20, 181-200, doi:10.1146/annurev-genom-091416-035517 (2019)
4. Lange, L. A. et al. Hum Mol Genet 19, 2050-2058, doi:10.1093/hmg/ddq062 (2010
5. Ojiaku, C. A., Yoo, E. J. & Panettieri, R. A., Jr. Am J Respir Cell Mol Biol 56, 432-442, doi:10.1165/rcmb.2016-0307TR (2017).

UNC CAROLINA POPULATION CENTER

Cebu Longitudinal Health and Nutrition Survey

UNIVERSITY of SAN CARLOS
SCIENTIA • VIRTUS • DEVOTIO